R3.1 Proton transfer reactions

Calculations involving pH and [H+]

Learning outcomes

By the end of this section you should be able to:

- (R3.1.4) Perform calculations involving the logarithmic relationship between pH and [H^+].
- (R3.1.5) Recognise solutions as acidic, neutral and basic from the relative values of [H^+] and [OH^-].

You have probably heard the term pH before, perhaps in a previous science course or while reading some article. You might have heard statements that compare the acidity of acid rain to that of normal rain or tap water, using pH values. But what is pH, what does it measure and how is it calculated?

The pH scale

The pH scale is used to determine how acidic or basic an aqueous solution is. The pH of a substance is a measure of the concentration of hydrogen ions, [H^+], in solution.

Table 1. Equation to calculate pH.

Equation	Symbols	Units
$\mathrm{pH} = -\log[\mathrm{H^+(aq)}]$	pH = a measure of how acidic or basic a species is	No units
	$[\mathrm{H^+(aq)}]$ = concentration of aqueous hydrogen ions	Moles per cubic decin (mol dm^{-3})

This equation can also be rearranged and expressed as:

$$[\mathrm{H^+}] = 10^{-\mathrm{pH}}$$

Study skills

The equation for pH can be found in section 1 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/some-relevant-equations-id-45103/) of the IBDP chemistry data booklet.

Recall that in section S1.4.5a (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/molar-concentration-of-a-solution-id-44263/) we saw that square brackets are used to show the concentration of a substance. In this topic $[H^+]$ refers to the concentration of hydrogen ions in solution and $[OH^-]$ refers to the concentration of hydroxide ions in solution.

The pH scale is an inverse scale; the higher the concentration of H^+ ions in solution, $[H^+]$, the lower the pH value. Conversely, the lower the $[H^+]$, the higher the pH value. In terms of hydroxide ions, a higher concentration of OH^- ions, $[OH^-]$, corresponds to a higher pH value and a lower $[OH^-]$ corresponds to a lower pH value (**Figure 1**). Solutions commonly found in the school laboratory have a range of pH values from 0 (acidic) to 14 (alkaline). It should be noted that pH values below 0 and above 14 are possible.

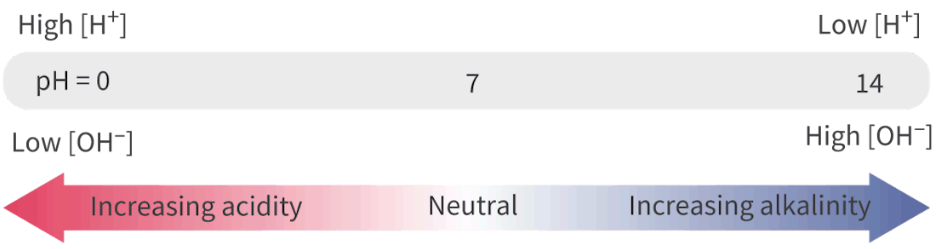

Figure 1. The pH scale is an inverse scale; the higher the $[H^+]$, the lower the pH value.

More information for figure 1

The image is a diagram of the pH scale, illustrating the relationship between hydrogen ions (H+) and hydroxide ions (OH-) in a solution. The scale is depicted as a horizontal bar ranging from pH 0 to 14. On the left end of the scale, at pH 0, it indicates a high concentration of hydrogen ions and low concentration of hydroxide ions, marked as 'High [H+] Low [OH-]'. On the right end, at pH 14, it shows a low concentration of hydrogen ions and high concentration of hydroxide ions, labeled as 'Low [H+] High [OH-]'. The middle of the scale, pH 7, represents a neutral point. Below the scale, an arrow extends from left to right, transitioning in color from red on the left (indicating increasing acidity) to blue on the right (indicating increasing alkalinity), with the word 'Neutral' in the center below pH 7.

[Generated by AI]

The relationship between $[H^+]$ and $[OH^-]$ in aqueous solutions is shown in **Figure 2.** Note that the diagram refers to the indicator litmus, which we will discuss in more detail in section R3.1.14-15 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/acid-base-indicators-hl-id-46062/).

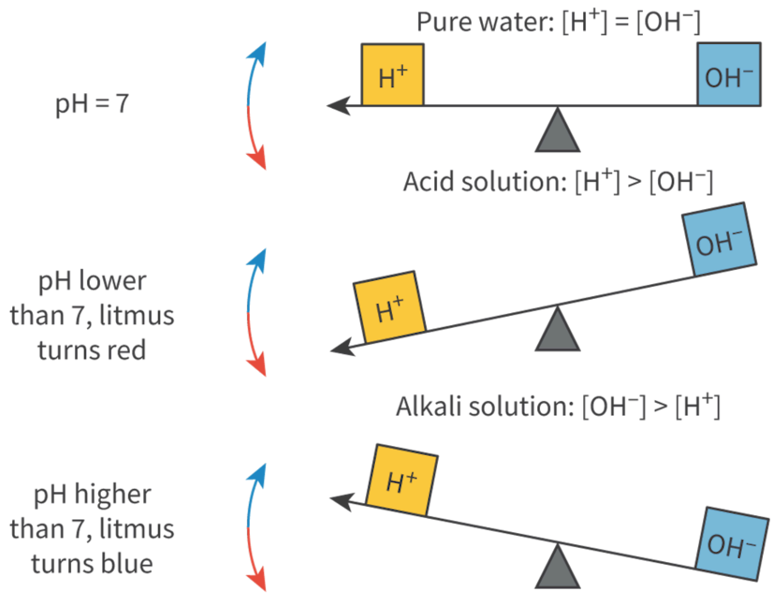

Figure 2. The balance between $[H^+]$ and $[OH^-]$ in aqueous solutions.

More information for figure 2

The diagram illustrates the balance between hydrogen ions (H+) and hydroxide ions (OH-) using a see-saw analogy. It consists of three scenarios:

1. At the top, the see-saw is level, representing neutral conditions where [H+] equals [OH-], typical of pure water. The text notes that the pH is 7.
2. The middle section shows the see-saw tilted to the left, indicating acidic conditions where [H+] is greater than [OH-]. This section notes that pH is lower than 7, and litmus turns red.
3. At the bottom, the see-saw tilts to the right, representing alkaline conditions where [OH-] is greater than [H+]. It is stated that pH is higher than 7, and litmus turns blue.

Each scenario is accompanied by rectangular labels representing H+ and OH-. Arrows beside the see-saws depict directionality correlated with the see-saw balance, indicating the shifts in acidity or alkalinity.

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If the concentration of hydrogen ions is equal to the concentration of hydroxide ions , the solution is neutral. Pure water is an example of a neutral substance because $[H^+] = [OH^-]$. If the $[H^+]$ is greater than the $[OH^-]$, the solution is acidic and if the $[OH^-]$ is greater than the $[H^+]$, the solution is basic, or alkaline. The relationship between $[H^+]$ and $[OH^-]$ is summarised in the **Table 1.**

Table 1. Acidic, basic or neutral?

	Acidic, basic or neutral	pH at 298 K
$[H^+] = [OH^-]$	neutral	7
$[H^+] > [OH^-]$	acidic	<7
$[OH^-] > [H^+]$	basic	>7

From the mathematical definition of pH, you should understand that this scale is logarithmic to the base 10. This means that a change of one pH unit corresponds to ten times the change in hydrogen ion concentration. For example, an aqueous solution with a pH = 3 is 10 times more acidic than an aqueous solution with a pH = 4; and 100 times (or 10 × 10) more acidic than an aqueous solution with a pH = 5.

Practical skills

Tool 3: Mathematics — Carry out calculations involving logarithmic functions, Use and interpret scientific notation

Remember:

- A logarithmic value such as pH should have the same number of decimal places as the least number of significant figures in the values used to calculate it.
- When in doubt with significant figures, it is always useful to convert the numbers to scientific notation.

Table 2 shows some values that emphasise the relationship between the concentration of hydrogen ions and pH. Not all the values are shown, but you should be able to work them out from the pattern.

Table 2. The relationship between $[H^+]$ concentration and pH.

pH	$[H^+(aq)]$ / mol dm^{-3}
0	1×10^0 or 1
1	1×10^{-1} or 0.1
2	1×10^{-2} or 0.01

pH	$[H^+(aq)]$ / mol dm^{-3}
6	1×10^{-6} or 0.000001
7	1×10^{-7} or 0.0000001
8	1×10^{-8} or 0.00000001
13	1×10^{-13} or 0.0000000000001
14	1×10^{-14} or 0.00000000000001

Table 2 demonstrates that if the pH of a solution is given the value x, then this corresponds to a hydrogen ion concentration of 10^{-x} mol dm^{-3}.

Study skills

A change in one pH unit represents a tenfold change in $[H^+]$. An increase of two pH units, for example, represents a 100 times change in $[H^+]$.

Hence, a solution with a pH = 1 has 10 times more hydrogen ions than a solution with pH = 2, and 100 times more than a pH = 3.

Every time that we want to state the strength of an acid we need to express the $[H^+]$ using scientific notation or write out a string of zeros. If, however, we use the base 10 logarithm (log10) of very large or very small numbers, we can avoid the use of scientific notation and it also allows us to compare the concentration of hydrogen ions in solution more easily.

Nature of Science

Aspect: Theories, Models

Why use a pH scale — Occam's razor.

The pH scale is an attempt to scale the relative acidity over a wide range of H^+ concentrations into a very simple number. It is an example of applying Occam's razor.

Occam's razor is the philosophical principle that the simplest scientific explanation which fits all known facts is the best one.

Sørensen's idea behind the pH scale was just to find a convenient way of expressing a very wide range of numbers in a simpler-to-write form. He used a logarithmic scale — it is a convenience linked to the reality of the concentration of hydrogen and hydroxide ions in pure water at 25 °C.

Scientists often use a logarithmic scale when there is a wide range of values to be expressed and considered. We have seen this when plotting the successive first ionisation energies of atoms and more generally on the Richter scale associated with measuring the strength of earthquakes.

Using such a scale is useful when the significance of a change in value lies in its relation to the value itself, rather than any absolute size. pH is defined as the negative of the logarithm of the concentration of hydrogen ions. The use of a logarithmic scale eases the written presentation of the figures, and also their graphical display. Otherwise, for instance, we would be writing the $[H^+]$ in neutral water as 0.0000001 mol dm^{-3} (or 10^{-7} mol dm^{-3}) — both of which are less easy on the eye than pH 7.0. The use of the term pH aids the discussion of values and also their graphical presentation.

pH calculations

From the definition of pH, it is possible to calculate the pH value of a solution from its $[H^+]$. For strong acids or bases, the $[H^+]$ or $[OH^-]$ in solution is equal to the initial concentration of the strong acid or base. This is because strong acids and bases completely dissociate in solution. Rearranging the equation for pH allows us to calculate the concentration of H^+ ions in a solution from its pH value:

$$\mathrm{pH} = -\log[\mathrm{H^+(aq)}]$$

$$[\mathrm{H^+(aq)}] = 10^{-\mathrm{pH}}$$

Worked example 1

Calculate the pH of 0.020 mol dm^{-3} nitric acid, HNO_3(aq).

Solution steps	Calculations
Step 1: Write out the values given in the question.	$[\mathrm{H^+}] = 0.020\ \mathrm{mol\ dm^{-3}}$ Note that $[\mathrm{H^+}]$ is equal to the initial conc because it is a strong acid that is comple
Step 2: Write out the equation.	$\mathrm{pH} = -\log_{10}[\mathrm{H^+}]$
Step 3: Substitute the values given.	$= 1.70$

Solution steps	Calculations
Step 4: State the answer with appropriate units and the number of significant figures used in rounding.	Since the digits to the left of the decimal exponential function, they do not represe Therefore, only the digits to the right of t measurement are significant.

Worked example 2

A solution has [H^+] of 1.0×10^{-8} mol dm^{-3}. Calculate its pH value.

Solution steps	Calcul
Step 1: Write out the values given in the question.	$[H^+] =$
Step 2: Write out the equation.	$pH =$
Step 3: Substitute the values given.	$= -\mathrm{lo}$
Step 4: State the answer with appropriate units and the number of significant figures used in rounding.	$= 8.0$ Recall units.

Worked example 3

Calculate the [H^+] of a solution with a pH value of 5.30.

Solution steps	
Step 1: Write out the values given in the question.	
Step 2: Write out the equation.	[
Step 3: Substitute the values given.	=

Solution steps	(
Step 4: State the answer with appropriate units and the number of significant figures used in rounding.	=

Acid-base indicators

In sections R3.1.8 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ph-curves-for-strong-acid-base-neutralisation-reactions-id-46057/) and R3.1.14–15 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/acid-base-indicators-hl-id-46062/), you will learn that an indicator changes colour depending on the pH of the solution. Two common indicators, litmus paper and universal indicator, are discussed below.

Litmus paper

Litmus paper is a common indicator used in the school laboratory. Like many indicators, it is made from a natural source, lichen. Litmus paper comes in two colours: red and blue. In an acidic solution, blue litmus changes to red and in an alkaline solution, red litmus changes to blue.

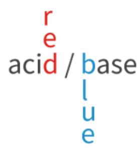

Figure 3. The colours of litmus paper in acidic and basic solutions.

More information for figure 3

The diagram illustrates the color changes of litmus paper based on the pH of the solution. The text "red" is in red color oriented vertically, representing the change of blue litmus paper to red in acidic conditions. Adjacent to it is the word "acid" in gray, centrally positioned. To the right, in blue, vertically oriented, is the text "blue," indicating the change of red litmus paper to blue in basic conditions. Below "blue" is the word "base" in gray. The diagram visually explains the basic and acidic changes in litmus paper.

[Generated by AI]

Other indicators give different colours in acidic and alkaline solutions. Some common examples are given in **Table 3**.

Table 3. Some common examples of acid-base indicators and their colours.

Indicator	Colour in acid	Colour in alkali
Litmus	red	blue
Methyl orange	red	yellow
Phenolphthalein	colourless	pink/purple

Although litmus is a very commonly used indicator, its usefulness is limited as it is only a qualitative indicator – it can detect if something is acidic or basic but it cannot determine the pH value. This is also true of the other indicators that just have a single colour change.

Universal indicator

Universal indicator is a quantitative indicator as it is actually a mixture of indicators that produces different colours depending on the pH of the solution. The mixture is designed so that the colours produced as the pH increases usually correspond to a ‘rainbow’ sequence as shown in **Figure 4**. The pH of aqueous solutions can be measured by using universal indicator as either a solution or in paper form.

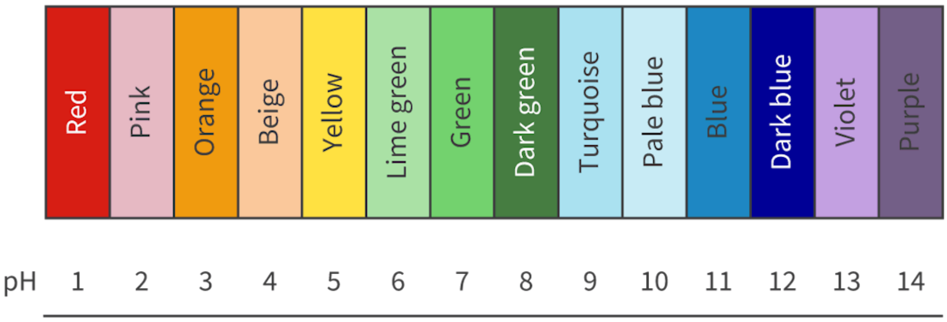

Figure 4. The pH scale and the universal indicator colours.

More information for figure 4

The image displays a pH scale ranging from 1 to 14. Beneath each pH value, there is a corresponding color which represents the universal indicator color for that pH level. Starting from pH 1, the sequence of colors includes red, pink, orange, beige, yellow, lime green, green, dark green, turquoise, pale blue, blue, dark blue, violet, and ends with purple at pH 14. The colors transition smoothly, creating a rainbow-like gradient that helps to identify pH levels based on color changes.

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Indicators can be used to measure the pH of a solution, by making use of the fact that their colour changes with pH. Visual comparison of the colour of a test solution with a standard colour chart allows us to measure the pH to the nearest whole number.

Figure 5. Reading the pH of a solution by matching to the colour chart.

Credit: Tetiana Garkusha, Getty Images

More information for figure 5

The image shows a hand holding two small vials containing colored solutions. The vials are placed over a pH color chart consisting of multiple colored squares. Each square on the chart is labeled with a pH value, ranging from acidic (red, orange) through neutral (yellow, green) to basic (dark green, blue). The vial on the left contains a yellow solution, while the vial on the right contains an orange solution. The hand is individually matching each vial to the closest color on the chart to determine the pH value of the solutions.

[Generated by AI]

Worked example 4

A substance has a pink colour with phenolphthalein and blue colour with litmus. Using **Table 3**, identify if the substance is an acid or a base.

The substance is a base since the colour of litmus with a base is blue and the colour of phenolphthalein is pink.

Practical skills

Tool 1: Experimental techniques — Measuring variables, pH of a solution

Tool 2: Technology — Applying technology to collect data

Can you think of a more accurate way to determine pH other than the use of universal indicator?

A pH probe and meter produces a more accurate method of measuring pH, provided they are correctly calibrated (**Figure 6**). The electrode of the pH meter is placed in the solution to be tested and a voltage is generated that is converted into a pH meter reading displayed on the screen. The pH meter is calibrated using buffers of known pH, usually of pH 4.0, 7.0 and 10.0.

For practical work you might use the pH probe or meter.

You will measure the pH of prepared solutions to determine their acidity, or determine pH at the equivalence point of a titration. The values of pH will indicate to you if a solution is acidic or basic and you would be able to derive information regarding the substances that are in excess or if a reaction is complete or not.

Figure 6. A pH meter reads the pH of a solution electronically.

Credit: photongpix, Getty Images

Aspect: Experiments

The relative strengths of acids and bases can now be measured using improved instrumentation.

Alongside the development of theories of acid base behaviour, the means of measuring acidity were also being enhanced. The concept of pH was an important step in that it allowed comparisons to be made between solutions of different acids at the same concentration. While no absolute comparison of the strengths of acids could be made, this enabled the relative strengths of different acids to be compared.

Technical and theoretical considerations went hand-in-hand. In 1889, Nernst, a German chemist, provided a theoretical foundation for using electrode potential to measure the ionic concentration of a solution. Then Sørensen, in 1900, developed the pH scale. Not only that, but he also devised an early form of pH meter using a hydrogen electrode combined with a calomel reference electrode.

Such a device added to the methods for estimating pH that used various indicators, including litmus. Then, in 1934, pH measurement was revolutionised by the development of the modern pH meter by Arnold Beckman. This instrument was originally developed for citrus fruit growers in California to monitor fruit acidity during the production of pectin and citric acid.

The ionic product of water

Pure water is a very poor conductor of electricity, which means that it must have a low concentration of mobile ions responsible for the electrical conductivity of solutions. Water molecules do dissociate however, but only to a very small extent. This is known as the auto-ionisation of water. This reaction can be represented by the following equation:

$$H_2O(l) \rightleftharpoons H^+(aq) + OH^-(aq)$$

The equilibrium constant expression (K_c) for the above reaction is:

$$\begin{aligned} K_c &= K_w \\ &= [H^+][OH^-] \end{aligned}$$

which is known as the ionic product of water. The ionic product of water is the product of the $[H^+]$ and $[OH^-]$ in water at a particular temperature. At 298 K (25°C), pure water has $[H^+]$ and $[OH^-]$ of 1.00×10^{-7} mol dm^{-3}. At 298 K, the value of K_w for pure water can be calculated as follows:

$$\begin{aligned} K_w &= [H^+][OH^-] \\ &= 1.00 \times 10^{-7} \times 1.00 \times 10^{-7} \\ &= 1.00 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6} \end{aligned}$$

Study skills

The equation for K_w and its value at 298 K are given in the DP chemistry data booklet in section 1 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/some-relevant-equations-id-45103/).

Like any equilibrium expression seen in subtopic R2.3 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43482/), K_w is temperature-dependent, therefore, at a certain temperature the product of $[H^+]$ and $[OH^-]$ is a constant. This means that as $[H^+]$ of a solution increases, $[OH^-]$ decreases (and vice versa). A neutral solution is any which has $[H^+]$ equal to $[OH^-]$. Recall that acidic solutions have $[H^+] > [OH^-]$ and basic solutions have $[OH^-] > [H^+]$.

Making connections

Can you make a connection to Le Châtelier's principle and think why does the ionisation of water increase as temperature increases? (see subtopic R2.3 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43482/)).

Calculating [H+] and [OH−]

Using the value of the ionic product of water, the $[H^+]$ and $[OH^-]$ of a solution can be calculated.

Worked example 5

An aqueous solution has a pH of 3.20 at 298 K.

Determine the $[H^+]$ and $[OH^-]$ of the solution.

Solution steps	Calculations
Step 1: Write out the values given in the question.	$\text{pH} = 3.20$
Step 2: Write out the equation.	$[\text{H}^+] = 10^{-\text{pH}}$
Step 3: Substitute the values given.	$= 10^{-3.20}$ $= 6.31 \times 10^{-4}$ m
Using the $[H^+]$ in solution and the K_w, we can determine the $[OH^-]$ in	
Step 4: Write out the required values.	$[\text{H}^+] = 6.31 \times 10$ $K_\text{w} = 1.00 \times 10^{-1}$

Solution steps	Calculations
Step 5: Write out the equation.	$K_w = [H^+][OH^-]$
Step 6: Rearrange the equation.	$[OH^-] = \frac{K_w}{[H^+]}$
Step 7: Substitute the values given.	$[OH^-] = \frac{1.00 \times}{6.31 \times}$
Step 8: State the answer with appropriate units and the number of significant figures used in rounding.	$= 1.58\times10^{-11}$ m

Worked example 6

Determine the pH of a 0.084 mol dm^{-3} solution of lithium hydroxide, LiOH at 298 K.

First calculate the value of $[H^+]$.

Lithium hydroxide is a strong base, therefore, it is completely ionised.

Solution steps	Calculations
Step 1: Write out the values given in the question.	$[OH^-] = 0.084 \text{ mol dm}^{-3}$ $K_w = 1.00\times10^{-14} \text{ mol}^2 \text{ dm}^{-6}$
Step 2: Write out the equation.	$K_w = [H^+][OH^-]$
Step 3: Rearrange the equation.	$[H^+] = \frac{K_w}{[OH^-]}$
Step 4: Substitute the values given.	$[H^+] = \frac{1.00 \times 10^{-14}}{0.084}$ $= 1.19 \times 10^{-13}$ mol dm

Now calculate the pH of the solution.

Solution steps	Calculations
Step 5: Write out the required values.	$[\mathrm{H}^+] = 1.19\times10^{-13}\ \mathrm{mol\ dm}^{-}$
Step 6: Write out the equation.	$\mathrm{pH} = -\log[\mathrm{H}^+]$
Step 7: Substitute the values given.	$\mathrm{pH} = -\log(1.19\times10^{-13})$ $= 12.92$
Step 8: State the answer with appropriate units and the number of significant figures used in rounding.	$= 13\ (2\ \mathrm{s.f.})$

Use the flashcards in **Interactive 1** to specify if the solution is acidic or basic based on the information given.

Specify if the solution is acidic or basic based on the given information.

pH is 3

Turn

Card 1 of 4

Interactive 1. Review Activity for Acidic Basic Determination Based on given Values.

Try the activity in the next section to check your understanding of pH.